

ONLINE RESULT PUBLISHING
WEBSITE AI23431- WEBTECHNOLOGY
AND MOBILE APPLICATION
MINI-PROJECT REPORT

SUBMITTED BY

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

RAJALAKSHMI ENGINEERING COLLEGE

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CHENNAI-602 105

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RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

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Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

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DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with an essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

CO-PO-PSO Mapping

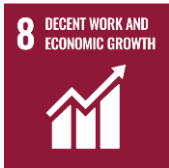
CO	PO 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	PS O 1	PS O 2	PS O 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3:

Substantial (High) No correlation: “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
 SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
 SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
 SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
 SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

This project is an Online Courier Management System designed to simplify the process of booking, tracking, and managing courier deliveries. It provides a convenient platform where customers can book shipments and track their parcels, administrators can manage delivery records, and courier staff can update shipment statuses efficiently.

The application allows administrators to create accounts, log in, and manage courier entries. Administrators can add new shipments with details such as sender information, receiver address, parcel weight, courier type, tracking number, and delivery status, as well as view complete delivery records. A tracking feature is also included, enabling customers to retrieve their shipment status by entering their tracking number or booking ID.

The system is developed using Java, JSP, and MySQL, and follows a structured approach to keep different parts of the application organized. Basic security features such as user authentication, role-based access control, and safe database operations are implemented to ensure reliability and data integrity.

Overall, the project demonstrates how web technologies can be used to build a practical and user-friendly platform for courier booking management and real-time shipment tracking.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JSP	Java Server Pages
JDBC	Java Database Connectivity
SQL	Structured Query Language
DBMS	Database Management System
UI	User Interface
UX	User Experience
HTTP	HyperText Transfer Protocol
OTP	One Time Password
API	Application Programming Interface
PDF	Portable Document Format

CHAPTER 1

INTRODUCTION

1.1 General

In today's digital era, the internet has become a primary medium for accessing information and services. The logistics and courier industry is no exception, as more businesses prefer to manage shipments and deliveries online rather than relying on traditional manual methods such as paper-based booking forms or physical tracking logs. Online platforms provide customers with quick access to their shipment status, making the process of courier management more efficient, transparent, and convenient.

This project focuses on the development of an Online Courier Management System that allows customers to book courier shipments, track their parcels in real time, and view delivery details. The system is designed to be user-friendly and accessible, enabling administrators to manage courier bookings efficiently and customers to retrieve their shipment status instantly by entering their tracking number or booking ID.

1.2 Need for Online Courier Management System

Traditional methods of courier management are often time-consuming, less transparent, and prone to errors. Customers usually have to visit courier offices in person or make phone calls to book shipments or inquire about delivery status, which can lead to delays and inconvenience. Similarly, administrators may find it difficult to manage and track a large number of shipments and delivery records efficiently.

An online courier management system addresses these challenges by providing a centralized platform where bookings can be made once and tracked by multiple users simultaneously. It reduces paperwork, improves accessibility, and allows customers to check their shipment status from anywhere at any time. Additionally, it ensures that delivery information is updated in real time, making the process more reliable, transparent, and efficient.

1.3 Scope of the System

The system is designed to serve as a comprehensive platform for courier booking and shipment management. It allows administrators to log in, add new courier entries with relevant details such as sender name, receiver name, address, contact number, parcel weight,

courier type, tracking number, and delivery status. The system also provides a tracking feature to help customers find their shipment details quickly by entering their unique tracking number.

Furthermore, customers can view complete shipment information including sender and receiver details, parcel weight, estimated delivery date, current delivery status, and courier type. The system is intended for courier companies, logistics firms, and e-commerce businesses, and can be extended in the future with additional features such as real-time GPS tracking, SMS/email notifications, payment integration, PDF invoice generation, and mobile application support.

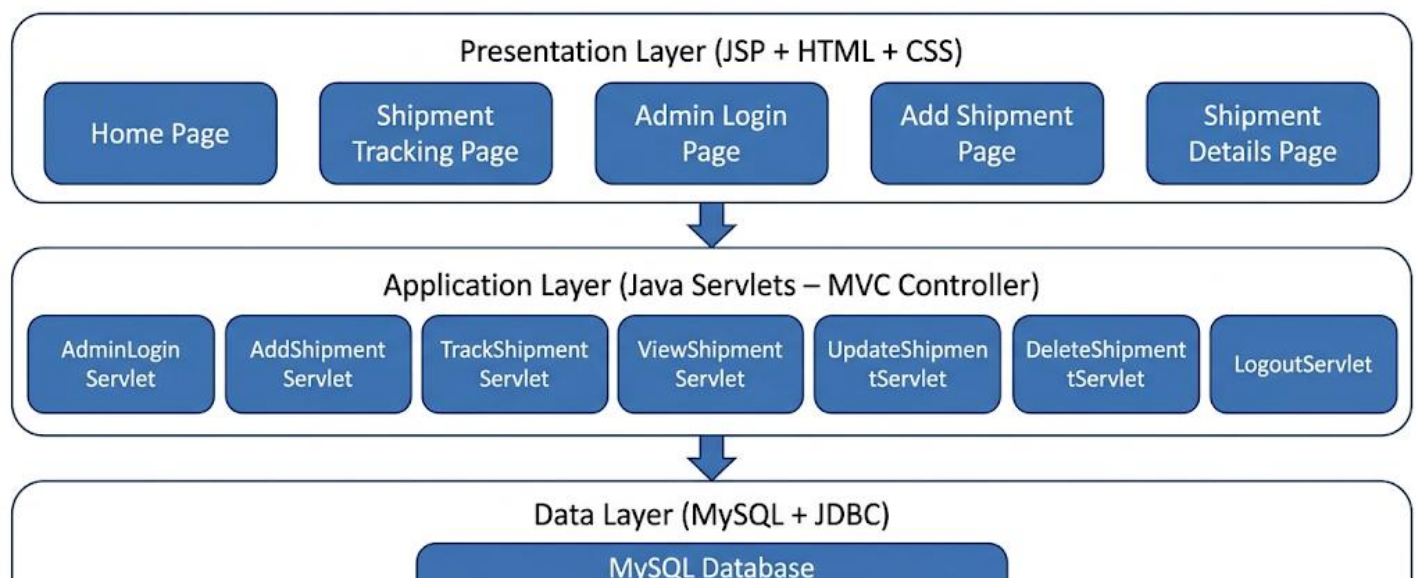
1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the Online Courier Management System.

- Chapter 1 provides the introduction to the project, including the general overview, need for the system, and scope of the project.
- Chapter 2 presents the literature survey, which discusses existing courier management platforms and related work.
- Chapter 3 explains the problem formulation and objectives of the proposed system.
- Chapter 4 describes the system design, including system architecture, database design, and system requirements.
- Chapter 5 discusses the implementation of the system and explains the different modules.
- Chapter 6 presents the results and discussion, including system outputs and performance evaluation.
- Chapter 7 concludes the project and suggests possible future enhancements.

1.5 System architecture diagram:

System Architecture – Online Courier Management System



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey explores existing systems and approaches used in online courier management and logistics platforms. With the rapid growth of internet usage and digital commerce, courier companies have increasingly adopted online solutions for shipment booking, tracking, and delivery management. This section examines how current platforms function, the technologies they use, and the challenges they face. Understanding these aspects helps in designing a system that is both efficient and user-friendly while addressing the limitations of existing solutions.

2.2 Existing Courier Management Platforms

Modern courier management platforms provide customers and administrators with the ability to book, track, and manage shipments digitally. These systems typically offer features such as tracking number-based search, sender and receiver information display, real-time delivery status updates, estimated delivery date calculation, and downloadable shipment receipts. Administrators can add bookings individually or in bulk and manage delivery records through centralized dashboards.

While these platforms are effective and widely used, they often come with certain limitations. Many systems are designed for large-scale enterprise use, making them complex and sometimes difficult for smaller courier businesses to implement. In addition, some platforms lack proper security measures, making customer data vulnerable to unauthorized access. Excessive features, slow loading times, or incomplete shipment information can also affect the overall user experience and trust in the system.

2.3 Related Work

Previous studies and projects in the domain of web-based courier management systems have focused on improving usability, security, efficiency, and real-time tracking. Researchers have explored the use of dynamic web technologies to create interactive user interfaces and faster data retrieval systems. Database-driven applications are commonly used to manage large volumes of shipment and customer data, ensuring that users can access updated and accurate information instantly.

Security and user authentication are also key areas of focus in related work, with systems implementing role-based access control, encrypted passwords, and session management to protect sensitive courier data. Some solutions incorporate advanced features such as live

GPS tracking, automated delivery notifications via email or SMS, payment gateway integration, performance dashboards, and integration with third-party logistics providers. However, these features often increase system complexity and may not be necessary for all businesses, especially smaller courier companies with limited technical resources.

2.4 Summary

The analysis of existing platforms and related work highlights the importance of providing a balanced system that combines functionality with simplicity. While many advanced systems offer a wide range of features, there is still a need for applications that are easy to use, secure, efficient, and focused on essential functionalities.

Based on this study, the proposed system aims to deliver a streamlined courier management platform that enables administrators to manage shipments and customers to track their parcels with ease. The focus is on creating a reliable, transparent, and user-friendly application that meets core requirements without unnecessary complexity.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In the current logistics environment, courier booking and shipment tracking is often carried out through traditional methods such as physical receipts, manual registers, or phone-based inquiries. These approaches can be time-consuming, inefficient, lack transparency, and are prone to errors or delays. Customers may struggle to access their shipment status promptly, while administrators face difficulties in managing and tracking a large number of deliveries efficiently.

Existing online platforms, although feature-rich, are often complex and may include unnecessary functionalities for basic business needs. They can also lack proper security measures, putting sensitive customer and shipment data at risk. Additionally, many systems do not provide a simple and centralized interface for courier booking and shipment retrieval. Therefore, there is a need for a streamlined and user-friendly web-based system that allows administrators to easily manage courier entries and enables customers to track their shipments quickly and securely using their tracking number or booking ID.

3.2 Objectives of the System

The main objective of the system is to develop a web-based application that simplifies the process of courier booking and shipment tracking. The specific objectives are as follows:

- To design and develop a user-friendly interface for courier booking and shipment tracking
- To enable customers to search and view their shipment status using tracking number or booking ID

- To provide functionality for administrators to add and manage courier shipment records
- To allow customers to view detailed shipment information including sender/receiver details and delivery status
- To implement a secure login and authentication system for administrators using session management
- To ensure safe and efficient database operations using proper connectivity techniques
- To calculate and display estimated delivery dates and overall shipment progress automatically
- To create a structured and maintainable system using appropriate design patterns

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system improves efficiency, reliability, and ease of maintenance.

The Online Courier Management System is designed as a web-based application that allows administrators to manage shipments and customers to track their parcels. It provides features such as admin authentication, courier booking entry, shipment tracking, and detailed delivery information display. The system follows a client-server architecture where users interact through web pages, while the backend processes requests and communicates with the database.

4.2 System Architecture

The system follows a three-layer architecture consisting of the presentation layer, application layer, and data layer.

Presentation Layer

This layer represents the user interface where users interact with the system. It includes web pages such as the homepage, shipment tracking page, delivery details page, admin login page, and booking entry page. These pages are developed using JSP, HTML, and CSS, and are responsible for displaying data to users and collecting input.

Application Layer

This layer handles the core logic of the system. It processes user requests, manages sessions,

performs validation, calculates delivery estimates, and controls the flow of data between the presentation layer and the database.

The backend is implemented using Java Servlets, where each servlet is responsible for a specific functionality:

- AdminLoginServlet → Handles administrator authentication
- AddShipmentServlet → Adds new courier booking records
- TrackShipmentServlet → Tracks shipment by tracking number
- ViewShipmentServlet → Displays detailed delivery information
- UpdateShipmentServlet → Updates existing shipment records
- DeleteShipmentServlet → Deletes shipment entries
- LogoutServlet → Manages admin logout

This layer ensures proper execution of all operations and maintains system functionality.

Data Layer

This layer is responsible for storing and managing all application data. The system uses a MySQL database connected through JDBC for data operations.

The database consists of three main tables:

- admin → Stores administrator credentials (admin_id, username, password)
- customers → Stores customer information (customer_id, name, contact, email, address)
- shipments → Stores courier shipment records (shipment_id, tracking_number, sender_id, receiver_name, receiver_address, weight, courier_type, status, booking_date, delivery_date)

4.3 Database Design

The database is designed to efficiently store and manage data related to administrators, customers, and shipments. Each table is structured with appropriate fields and relationships.

- The admin table stores login credentials for authorized users
- The customers table stores registered customer information
- The shipments table stores shipment details and is linked to customers through a foreign key

Relationships between tables ensure data consistency and integrity. For example, each shipment entry is associated with a specific customer, and multiple shipments can exist for the same customer across different booking dates and courier types.

Key Fields in Shipments Table:

- shipment_id (Primary Key)
- customer_id (Foreign Key referencing customers table)
- tracking_number
- receiver_name

- receiver_address
- parcel_weight
- courier_type
- delivery_status
- booking_date
- estimated_delivery_date

4.4 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.4.1 Hardware Requirements

The system does not require high-end hardware and can run on standard computers.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection: Required

4.4.2 Software Requirements

The system is developed using commonly available tools and technologies.

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JSP
- Backend Technology: Java Servlets
- Database: MySQL
- Database Connectivity: JDBC
- Web Server: Apache Tomcat 9.0
- Development Environment: Eclipse IDE for Enterprise Java
- Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The system is developed using a structured approach based on the Model-View-Controller (MVC) architecture. This methodology helps in separating the user interface, business logic, and data handling, making the system easier to develop and maintain.

- **Model:** Represents the data and business logic using Java classes such as Shipment, Customer, and Admin.
- **View:** Consists of JSP pages that display data to users and collect input.
- **Controller:** Implemented using Java Servlets, which handle user requests, process data, and control the flow between the view and model.

The development process involves designing the database, creating model classes, implementing servlets for different functionalities, and developing JSP pages for user interaction. JDBC is used to connect the application with the MySQL database.

5.2 Module Descriptions

The system is divided into several modules, each responsible for a specific functionality.

5.2.1 Admin Module

This module handles administrator authentication and session management.

- Administrators log in using valid username and password
- Session management is implemented using HttpSession
- Only authenticated admins can access shipment management features
- Admin credentials are stored securely in the database
- Logout functionality clears the session

5.2.2 Shipment Management Module

This module allows administrators to add, update, and delete courier shipment records.

- Admins can add new shipments with details such as tracking number, sender name, receiver name, address, parcel weight, and courier type

- Shipment entries are stored in the database with automatic delivery status tracking
- Admins can update existing shipment records
- Admins can delete incorrect or outdated shipment entries
- All operations maintain data integrity

5.2.3 Tracking Module

This module provides functionality for customers to track their shipments.

- Customers enter their tracking number or booking ID
- The system performs a search using SQL queries
- Matching shipment records are retrieved from the database
- If no shipment is found, an appropriate message is displayed
- Helps customers quickly access their delivery status

5.2.4 Shipment Details Module

This module displays complete shipment information to customers.

- Shows full shipment details including sender and receiver information
- Displays parcel weight, courier type, booking date, and estimated delivery date
- Presents current delivery status in a structured and readable format
- Allows customers to view complete delivery progress
- Improves transparency and customer satisfaction

5.2.5 Customer Verification Module

This module ensures that only valid customer data is entered.

- Validates tracking number format and uniqueness
- Checks for duplicate entries before adding shipments
- Ensures all required fields are filled
- Prevents invalid or incomplete data submission
- Maintains database accuracy

5.2.6 Login and Authentication Module

This module ensures secure access to the administrative panel.

- Validates admin credentials during login
- Uses session management to maintain admin state
- Prevents unauthorized access to shipment management features
- Handles login and logout operations securely
- Implements password protection

5.2.7 Delete/Update Shipment Module

This module allows administrators to modify or remove courier shipment entries.

- Admins can update delivery status, receiver details, or other shipment information
- Admins can delete incorrect shipment entries
- Confirmation prompts prevent accidental deletions
- Maintains an audit trail of changes
- Ensures data consistency

CHAPTER 6

RESULTS AND DISCUSSION

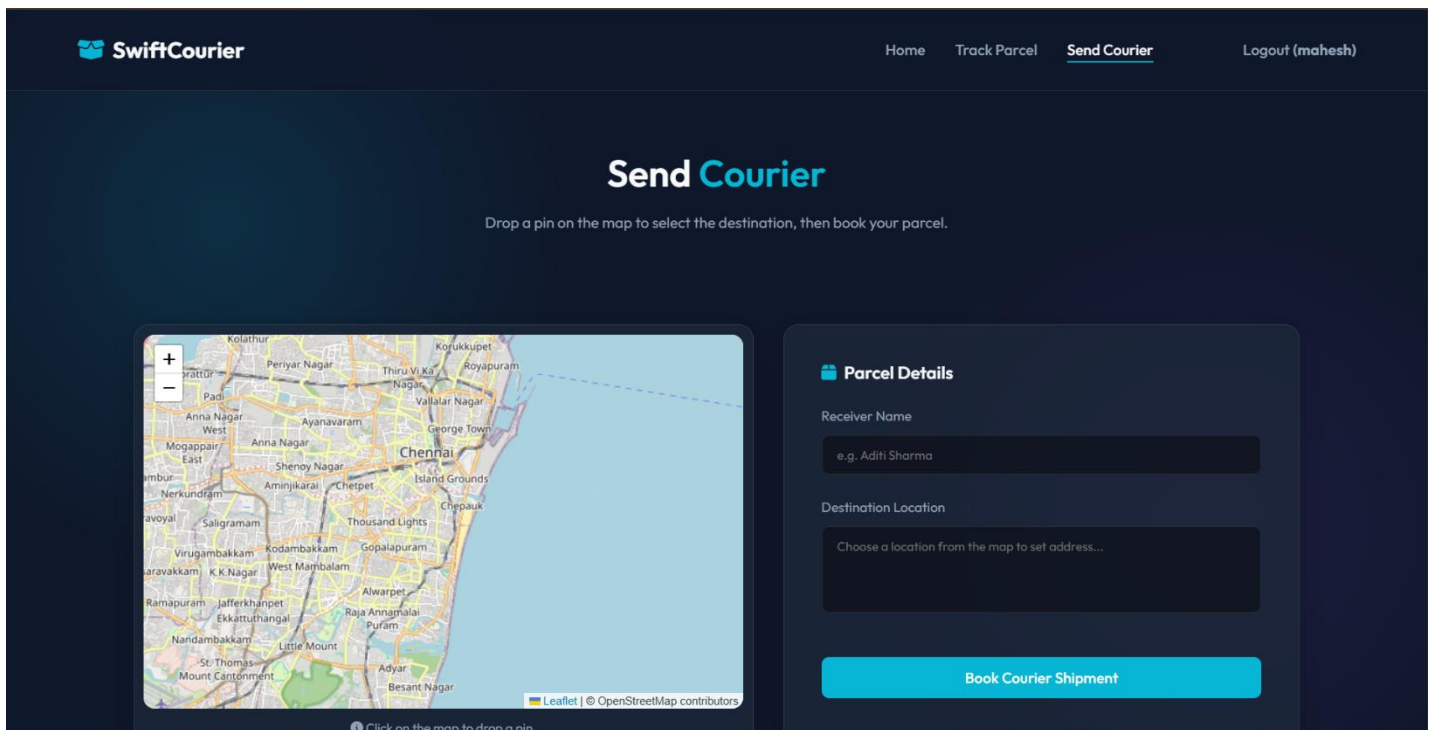
6.1 System Output Screens

The developed system was tested by executing all major functionalities, and the outputs were verified through different user interactions. The application successfully displays dynamic content retrieved from the database and responds correctly to user inputs.

The main output screens of the system include:

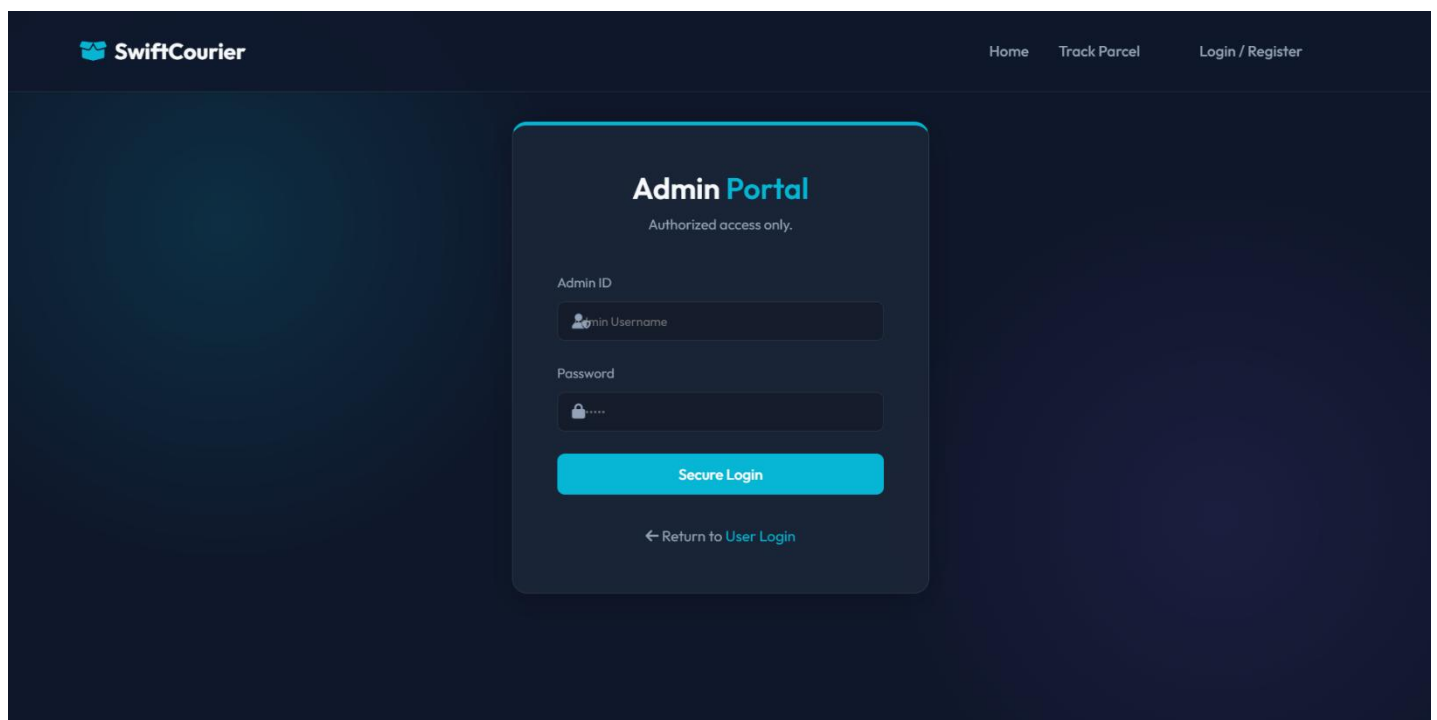
Customer Homepage (Shipment Tracking):

Displays a tracking interface where customers can enter their tracking number to retrieve shipment status. The page is simple and intuitive, with clear instructions.



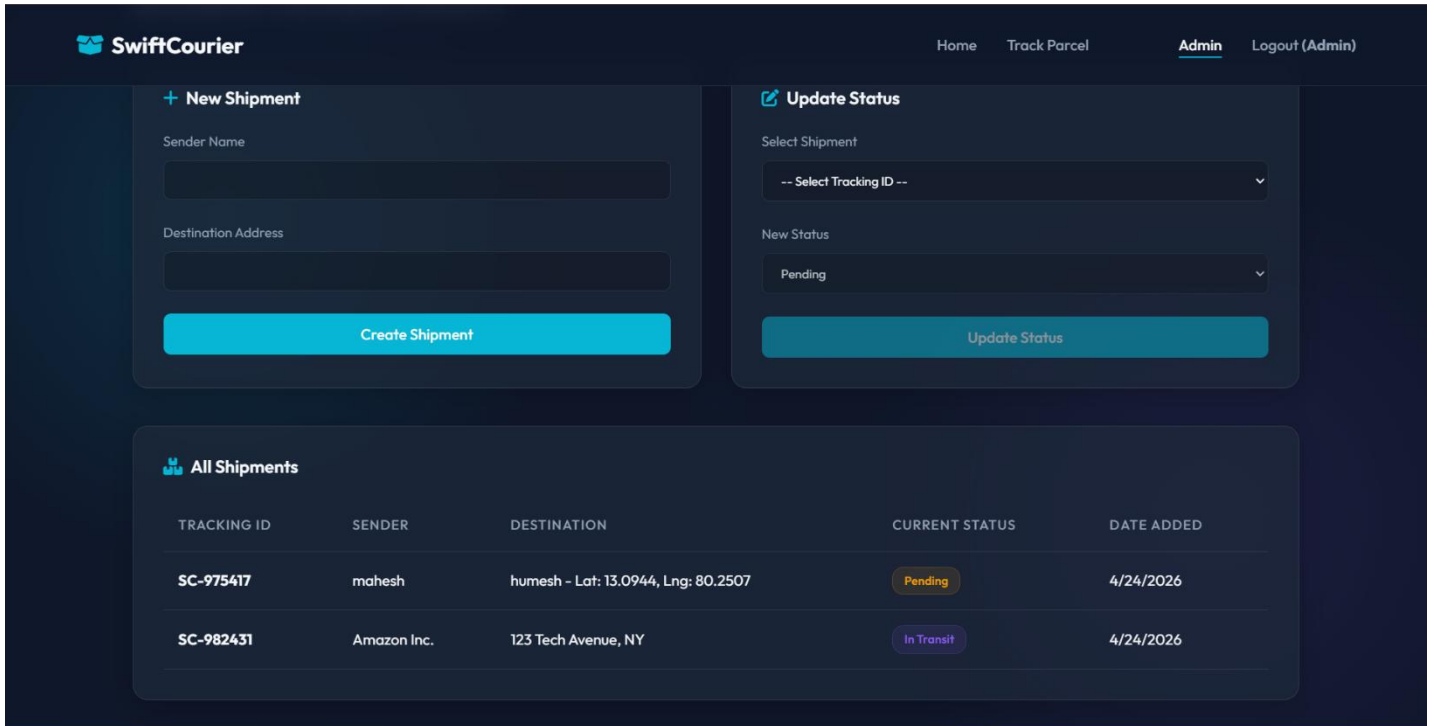
Admin Login Page:

Allows authorized administrators to log in using their username and password. Proper validation messages are displayed for incorrect credentials.



Admin Dashboard:

Displays options for adding, updating, viewing, and deleting shipment records. Provides easy navigation to all administrative functions.

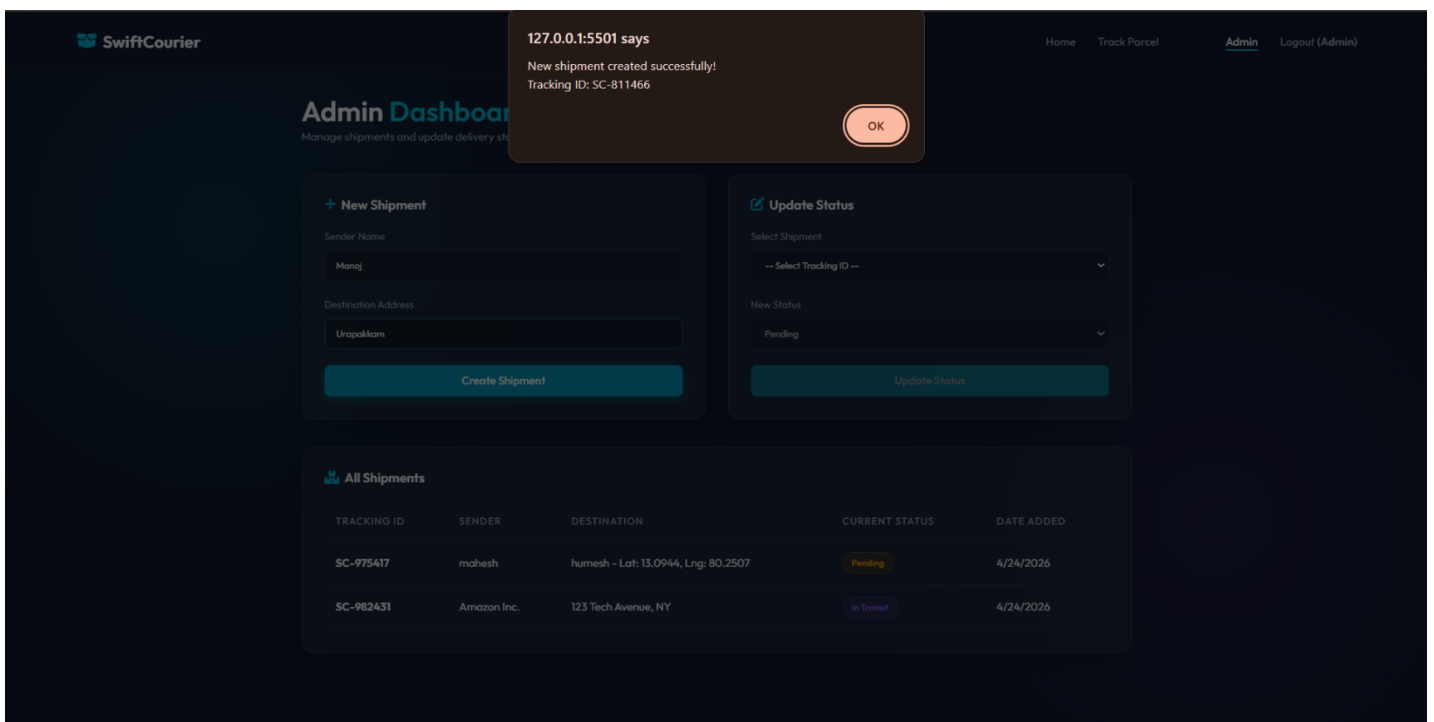


The screenshot shows the SwiftCourier Admin Dashboard. At the top, there's a navigation bar with 'Home', 'Track Parcel', 'Admin' (active), and 'Logout (Admin)'. The main content area is divided into three sections. The top left section, titled '+ New Shipment', has input fields for 'Sender Name' and 'Destination Address', and a 'Create Shipment' button. The top right section, titled 'Update Status', has a 'Select Shipment' dropdown (currently showing '-- Select Tracking ID --'), a 'New Status' dropdown (currently showing 'Pending'), and an 'Update Status' button. The bottom section, titled 'All Shipments', contains a table with the following data:

TRACKING ID	SENDER	DESTINATION	CURRENT STATUS	DATE ADDED
SC-975417	maresh	humesh - Lat: 13.0944, Lng: 80.2507	Pending	4/24/2026
SC-982431	Amazon Inc.	123 Tech Avenue, NY	In Transit	4/24/2026

Add Shipment Page:

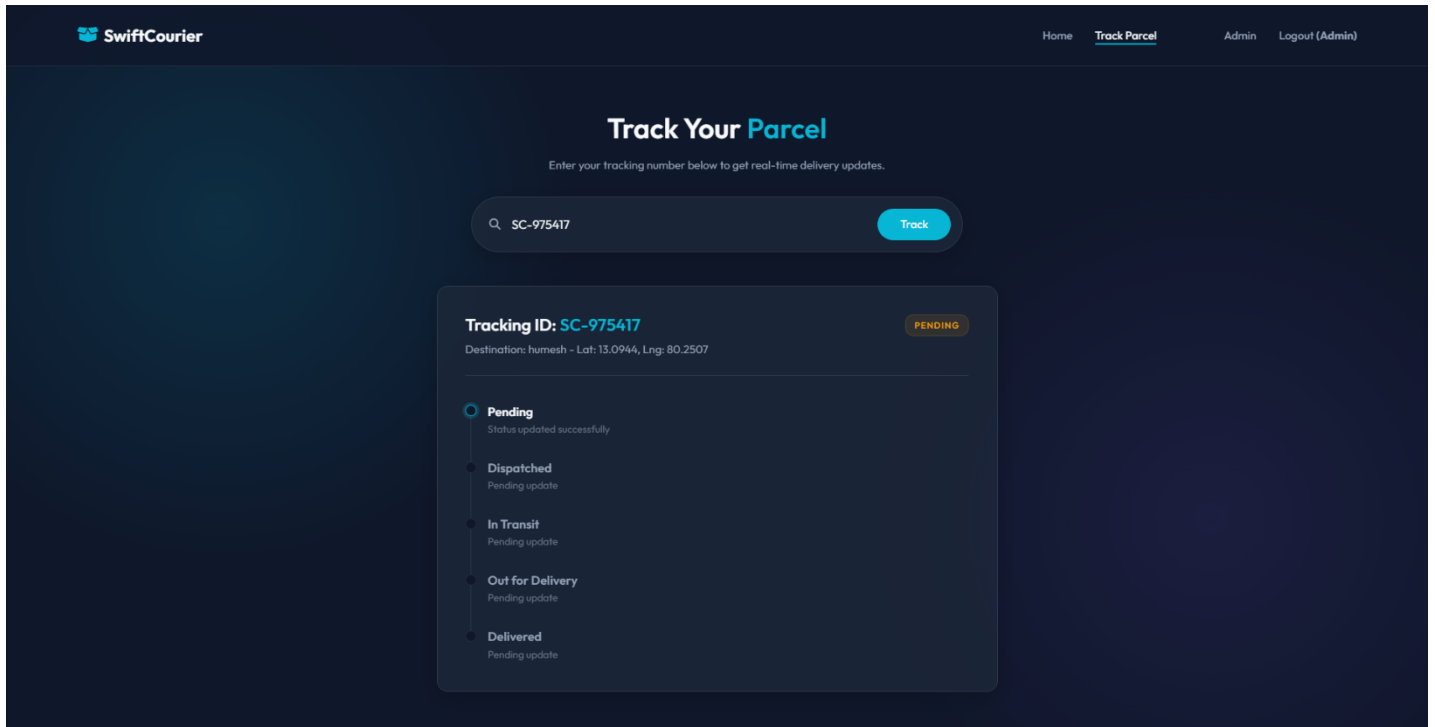
Enables administrators to enter courier booking details including sender name, receiver name, address, parcel weight, courier type, and tracking number. Includes validation for all input fields.



This screenshot shows the same SwiftCourier Admin Dashboard as the previous one, but with a success message overlay. The message box at the top center reads: '127.0.0.1:5501 says New shipment created successfully! Tracking ID: SC-811466'. Below the message is an 'OK' button. The dashboard layout and data remain the same as in the previous screenshot.

Track Shipment Page:

Shows the customer's complete shipment details after a successful tracking number search. Displays sender and receiver details, delivery status, and estimated delivery date in a formatted layout.



Shipment Details Page:

Presents detailed information including customer name, tracking number, parcel weight, courier type, booking date, and current delivery status.

Delete Confirmation Page:

Prompts administrators to confirm before deleting a shipment entry, preventing accidental data loss.

Update Shipment Page:

Allows administrators to modify existing shipment entries. Pre-fills current data and allows editing of delivery status and other relevant fields.

The screenshot displays the SwiftCourier Admin Dashboard. At the top, there's a navigation bar with 'Home', 'Track Parcel', 'Admin' (active), and 'Logout (Admin)'. The main heading is 'Admin Dashboard' with a subtitle 'Manage shipments and update delivery statuses globally.' Below this, there are three main sections:

- + New Shipment:** A form with 'Sender Name' and 'Destination Address' input fields, and a 'Create Shipment' button.
- Update Status:** A form with a 'Select Shipment' dropdown (showing 'SC-975417 (Pending)') and a 'New Status' dropdown menu. The menu is open, showing options: 'Pending', 'Dispatched', 'In Transit', 'Out for Delivery' (highlighted), and 'Delivered'.
- All Shipments:** A table listing existing shipments.

TRACKING ID	SENDER	DESTINATION	CURRENT STATUS	DATE ADDED
SC-811466	Manoj	Urapakkam	Pending	5/12/2026
SC-975417	mahesh	humesb - Lat: 13.0944, Lng: 80.2507	Pending	4/24/2026
SC-982431	Amazon Inc.	123 Tech Avenue, NY	In Transit	4/24/2026

These output screens demonstrate that the system is functioning correctly and providing the intended features with accuracy and clarity.

6.2 Performance Evaluation

The system was evaluated based on responsiveness, accuracy, and usability.

- **Response Time:** The system responds quickly to user requests such as tracking shipments, adding new bookings, and displaying delivery details. Average response time is under 2 seconds for most operations.
- **Data Accuracy:** All data retrieved from the database is displayed correctly without inconsistencies. Delivery status, estimated dates, and customer information are accurately maintained.
- **Security:** The system implements session-based authentication for administrators, preventing unauthorized access. Password fields are protected, and SQL injection vulnerabilities are minimized through prepared statements.
- **Usability:** The user interface is simple, clean, and easy to navigate. Customers can track shipments without confusion, and administrators can manage booking data efficiently.
- **Efficiency:** The use of JDBC and optimized SQL queries ensures efficient data handling and retrieval. The database is properly indexed for faster tracking searches.

Overall, the system performs well for small to medium-scale courier businesses and provides a smooth, reliable user experience.

6.3 System Testing

The system was tested using different scenarios to ensure reliability and correctness.

- **Functional Testing:** All features such as admin login, shipment booking, customer tracking, shipment display, update, and delete were tested and found to be working as expected.
- **Authentication Testing:** Admin login and session management were tested to ensure secure access. Unauthorized users cannot access administrative functions.
- **Validation Testing:** Input fields were tested for correct validation, ensuring that invalid, empty, or incorrect data formats are not accepted. Tracking number uniqueness is enforced.
- **Database Testing:** Data insertion, retrieval, update, and deletion operations were verified to ensure proper interaction with the database. Foreign key constraints and data integrity were validated.
- **Error Handling:** The system handles errors gracefully and displays appropriate messages to the user. For example, "No shipment found" when a tracking number does not exist.

The testing results confirm that the system is stable, reliable, secure, and ready for deployment in a courier management environment.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Online Courier Management System was successfully designed and developed to simplify the process of booking, tracking, and managing courier shipments. The application provides a user-friendly interface that allows customers to track their parcels quickly by entering their tracking number, and enables administrators to manage shipment records efficiently.

The system effectively integrates frontend and backend technologies using Java Servlets, JSP, and MySQL. It follows a structured MVC architecture that ensures clear separation of components, making the application easy to understand, maintain, and extend. Key features such as administrator authentication, shipment management, delivery status tracking, customer shipment search, and secure database interaction have been implemented successfully.

The project demonstrates the practical application of web development concepts and database management in the logistics sector. It provides a reliable, transparent, and efficient platform for handling courier bookings, eliminating the need for manual record-keeping and reducing delays. Overall, the system meets its objectives and performs efficiently for small to medium-scale courier businesses.

7.2 Future Enhancements

Although the system meets the current requirements, it can be further improved by adding more advanced features in the future:

- **Real-Time GPS Tracking:** Integrating live GPS location tracking to allow customers to view the exact real-time location of their parcel during transit.
- **Shipment Analytics Dashboard:** Displaying delivery performance graphs, charts, region-wise statistics, on-time delivery rates, and monthly shipment comparisons for better business insights.
- **PDF Invoice and Receipt Support:** Allowing customers to download or print their shipment receipts and invoices in PDF format for record keeping and business purposes.
- **Mobile Application:** Developing a mobile version of the system (Android/iOS) for improved accessibility and convenience for both customers and delivery staff.
- **Bulk Shipment Upload:** Providing an option for administrators to upload multiple shipment entries in bulk using Excel or CSV files, reducing manual data entry time.
- **Customer Portal:** Creating a dedicated customer login where they can view all past and current shipments, track delivery history, and download shipment certificates.
- **Multi-level Access Control:** Implementing different roles such as Super Admin, Courier Staff, and Data Entry Operator with specific permissions for each role.
- **Shipment Verification System:** Adding QR codes or unique verification codes on shipment receipts for authenticity verification by receivers and external organizations.
- **Email and SMS Notifications:** Sending automatic notifications to customers when their shipment status is updated, including estimated delivery time and a link to track their parcel online.
- **Payment Gateway Integration:** Adding online payment support to allow customers to pay for their courier bookings directly through the system using debit/credit cards or UPI.
- **Improved Security:** Enhancing security with password encryption (hashing), two-factor authentication, and CAPTCHA on login pages.

These enhancements can make the system more scalable, interactive, and suitable for real-world deployment on a larger scale.

REFERENCES

- Oracle Corporation, *Java Servlet Technology Documentation*. Available: <https://docs.oracle.com/javaee/>
- Oracle Corporation, *JSP (Java Server Pages) Documentation*. Available: <https://docs.oracle.com/javaee/>
- Oracle Corporation, *JDBC API Documentation*. Available: <https://docs.oracle.com/javase/>
- Apache Software Foundation, *Apache Tomcat Documentation*. Available:

<https://tomcat.apache.org/>

- MySQL Documentation Team, *MySQL Reference Manual*. Available: <https://dev.mysql.com/doc/>
- MDN Web Docs, *HTML, CSS, and Web Development Guides*. Available: <https://developer.mozilla.org/>
- W3Schools, *Web Development Tutorials*. Available: <https://www.w3schools.com/>
- GeeksforGeeks, *Java and Web Development Articles*. Available: <https://www.geeksforgeeks.org/>
- K. Sierra and B. Bates, *Head First Servlets and JSP*, O'Reilly Media, 2nd Edition, 2008.
- E. Balagurusamy, *Programming with Java*, McGraw Hill Education, 6th Edition.

APPENDICES

Appendix-1: Sample Database Tables

Admin Table

Field Name	Data Type	Description
admin_id	INT (PK)	Unique admin ID
username	VARCHAR	Admin username
password	VARCHAR	Admin password
email	VARCHAR	Admin email

Customers Table

Field Name	Data Type	Description
customer_id	INT (PK)	Unique customer ID
name	VARCHAR	Customer full name
contact	VARCHAR (UNIQUE)	Customer phone number
email	VARCHAR	Customer email
address	VARCHAR	Customer address

Shipments Table

Field Name	Data Type	Description
shipment_id	INT (PK)	Unique shipment ID
customer_id	INT (FK)	Reference to customer
tracking_number	VARCHAR	Unique tracking number
receiver_name	VARCHAR	Receiver full name
receiver_address	VARCHAR	Delivery address
parcel_weight	DECIMAL	Weight in kg
courier_type	VARCHAR	Type (Express/Standard)
delivery_status	VARCHAR	Status (Booked/In Transit/Delivered)
booking_date	DATE	Date of booking
estimated_delivery_date	DATE	Expected delivery date

Appendix-2: Sample SQL Queries

-- Insert Admin

```
INSERT INTO admin (username, password, email)
VALUES ('admin', 'admin123', 'admin@courier.com');
```

-- Insert Customer

```
INSERT INTO customers (name, contact, email, address)
VALUES ('Mahesh K', '9876543210', 'mahesh@email.com', 'Chennai, Tamil Nadu');
```

-- Insert Shipment


```
INSERT INTO shipments (customer_id, tracking_number, receiver_name, receiver_address,
parcel_weight, courier_type, delivery_status, booking_date, estimated_delivery_date)
VALUES (1, 'TRK2024001', 'Raj Kumar', 'Bangalore, Karnataka',
2.5, 'Express', 'In Transit', '2024-01-10', '2024-01-12');
```

-- Track Shipment by Tracking Number

```
SELECT * FROM shipments WHERE tracking_number = 'TRK2024001';
```

-- Get Shipment with Customer Details

```
SELECT c.name, c.contact, s.tracking_number, s.receiver_name,
s.receiver_address, s.parcel_weight, s.courier_type, s.delivery_status,
s.booking_date, s.estimated_delivery_date
FROM customers c
JOIN shipments s ON c.customer_id = s.customer_id
WHERE s.tracking_number = 'TRK2024001';
```

-- Update Shipment Status

```
UPDATE shipments
SET delivery_status = 'Delivered'
WHERE shipment_id = 1;
```

-- Delete Shipment

```
DELETE FROM shipments WHERE shipment_id = 1;
```

Appendix-3: Key Servlet Example

@WebServlet("/trackShipment")

```
public class TrackShipmentServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
```

```
        String trackingNumber = request.getParameter("trackingNumber");
        Shipment shipment = null;
```

```
        Connection con = DBConnection.getConnection();
```

```

try {
    String query = "SELECT c.name, c.contact, s.tracking_number, " +
        "s.receiver_name, s.receiver_address, s.parcel_weight, " +
        "s.courier_type, s.delivery_status, s.booking_date, " +
        "s.estimated_delivery_date " +
        "FROM customers c JOIN shipments s ON c.customer_id = s.customer_id " +
        "WHERE s.tracking_number = ?";

    PreparedStatement pst = con.prepareStatement(query);
    pst.setString(1, trackingNumber);
    ResultSet rs = pst.executeQuery();

    if (rs.next()) {
        shipment = new Shipment();
        shipment.setSenderName(rs.getString("name"));
        shipment.setSenderContact(rs.getString("contact"));
        shipment.setTrackingNumber(rs.getString("tracking_number"));
        shipment.setReceiverName(rs.getString("receiver_name"));
        shipment.setReceiverAddress(rs.getString("receiver_address"));
        shipment.setParcelWeight(rs.getDouble("parcel_weight"));
        shipment.setCourierType(rs.getString("courier_type"));
        shipment.setDeliveryStatus(rs.getString("delivery_status"));
        shipment.setBookingDate(rs.getString("booking_date"));
        shipment.setEstimatedDeliveryDate(rs.getString("estimated_delivery_date"));
    } else {
        request.setAttribute("message", "No shipment found for this tracking number");
    }
} catch (Exception e) {
    e.printStackTrace();
}

request.setAttribute("shipment", shipment);
request.getRequestDispatcher("viewShipment.jsp").forward(request, response);
}
}

```

PSEUDOCODE

The following pseudocode represents the core functionalities of the Online Courier Management System.

Admin Login

BEGIN

INPUT username, password

CONNECT to database

FETCH admin where username and password match

IF admin exists THEN

 CREATE session

 STORE admin details in session

 REDIRECT to admin dashboard

ELSE

 DISPLAY "Invalid credentials"

ENDIF

CLOSE database connection

END

Add Shipment

BEGIN

CHECK if admin is logged in

INPUT sender_name, receiver_name, receiver_address, contact,
 parcel_weight, courier_type, tracking_number, booking_date

CONNECT to database

CALCULATE estimated_delivery_date based on courier_type:

 IF courier_type = "Express" THEN estimated_delivery_date = booking_date + 2 days

 ELSE IF courier_type = "Standard" THEN estimated_delivery_date = booking_date + 5 days

 ELSE estimated_delivery_date = booking_date + 7 days

SET delivery_status = "Booked"

CHECK if customer exists in customers table

IF NOT exists THEN

 INSERT customer details into customers table

ENDIF

INSERT shipment details into shipments table

DISPLAY "Shipment booked successfully"

CLOSE database connection

END

Track Shipment by Tracking Number

```
BEGIN INPUT tracking_number CONNECT to database EXECUTE query: SELECT
c.name, c.contact, s.tracking_number, s.receiver_name, s.receiver_address, s.parcel_weight,
s.courier_type, s.delivery_status, s.booking_date, s.estimated_delivery_date FROM
customers c JOIN shipments s ON c.customer_id = s.customer_id WHERE
s.tracking_number = tracking_number IF shipment found THEN STORE shipment details
DISPLAY sender name, receiver details, courier type DISPLAY parcel weight, booking
date, estimated delivery date DISPLAY current delivery status ELSE DISPLAY "No
shipment found for this tracking number" ENDIF CLOSE database connection END
```

View All Shipments (Admin)

BEGIN

CHECK if admin is logged in

CONNECT to database

FETCH all shipments ordered by booking_date DESC

STORE shipments in list

SEND list to JSP page

DISPLAY shipments in tabular format

CLOSE database connection

END

Update Shipment

BEGIN

CHECK if admin is logged in

GET shipment_id from request

CONNECT to database

INPUT new delivery_status

UPDATE shipments table:

 SET delivery_status = new_status

 WHERE shipment_id = shipment_id

DISPLAY "Shipment updated successfully"

CLOSE database connection

END

Delete Shipment

BEGIN

CHECK if admin is logged in

GET shipment_id from request

CONNECT to database

```
DELETE FROM shipments WHERE shipment_id = shipment_id
DISPLAY "Shipment deleted successfully"
CLOSE database connection
END

Calculate Delivery Status Progress
BEGIN
INPUT delivery_status
IF delivery_status = "Booked" THEN progress = 25%
ELSE IF delivery_status = "Picked Up" THEN progress = 50%
ELSE IF delivery_status = "In Transit" THEN progress = 75%
ELSE IF delivery_status = "Out for Delivery" THEN progress = 90%
ELSE IF delivery_status = "Delivered" THEN progress = 100%
ELSE progress = 0%
RETURN progress
END

Admin Logout
BEGIN
GET current session
INVALIDATE session
REDIRECT to login page
END
```

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